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SMALL CELL REACTORS WITH SHARED FORELINE AND PRESSURE CONDUIT

Abstract

In one embodiment, a processing system for semiconductor manufacturing, includes a chamber housing, a first process chamber, a second process chamber and a foreline disposed in the chamber housing and between the first process chamber and the second process chamber. The first and second process chambers are in the chamber housing and each includes a pump ring and a liner. The pump ring includes a port and baffle. The pump rings and liners each partially define a first and second process volume respectively. The foreline includes a first exhaust chamber fluidly coupled to the first process volume and a second exhaust chamber fluidly coupled to the second process volume.

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Background/Summary

BACKGROUND

Field

[0001] Embodiments of the present invention generally relate to components, apparatus, and systems for semiconductor manufacturing. More specifically, the present technology relates to processing chamber components and other semiconductor processing equipment.

Description of the Related Art

[0002] Integrated circuits are made possible by processes which produce intricately patterned material layers on substrate surfaces. Producing patterned material on a substrate requires controlled methods for forming and removing material. Precursors are often delivered to a processing region and distributed to uniformly deposit or etch material on the substrate. Many aspects of a processing chamber may impact process uniformity, such as uniformity of process conditions within a chamber, uniformity of flow through components, as well as other process and component parameters. Even minor discrepancies between substrate processing chambers may impact the formation or removal of material between similar substrates.

[0003] Thus, there is a need for improved systems and methods that can be used to produce high quality devices and structures with enhanced throughput and increased accuracy. These and other needs are addressed by the present technology.

SUMMARY

[0004] To be completed Embodiments herein include a processing system for semiconductor manufacturing. In one embodiment, a processing system for semiconductor manufacturing is provided. The processing system includes a chamber housing, a first process chamber, a second process chamber and a foreline disposed in the chamber housing and between the first process chamber and the second process chamber. The first process chamber is in the chamber housing and includes a first pump ring and a first liner. The first pump ring includes a first port and first baffle. The first pump ring and first liner partially define a first process volume. The second process chamber is in the chamber housing and includes a second pump ring and a second liner. The second pump ring includes a second port and second baffle. The second pump ring and second liner partially define a second process volume. The foreline is in the chamber housing and between the first process chamber and the second process chamber. The foreline includes a first exhaust chamber fluidly coupled to the first process volume and a second exhaust chamber fluidly coupled to the second process volume.

[0005] In another embodiment, a processing system for semiconductor manufacturing is provided. The processing system includes a chamber housing, a first process chamber, a second process chamber, a pressure sensor, and a foreline disposed in the chamber housing and between the first process chamber and the second process chamber. The first process chamber is disposed in the chamber housing and includes a first pump ring and a first liner. The first pump ring includes a first port, a first baffle, and a first pressure conduit disposed in the first pump ring. The first pump ring and first liner partially define a first process volume. The second process chamber is in the chamber housing and includes a second pump ring and a second liner. The second pump ring includes a second port, a second baffle, and a second pressure conduit disposed in the second pump ring. The second pump ring and second liner partially define a second process volume. The pressure sensor is in fluid communication with the first process volume by the first pressure conduit, the chamber housing disposed at least partially between the pressure sensor and the first pressure conduit of the first pump ring. The foreline includes a first exhaust chamber fluidly coupled to the first process volume and a second exhaust chamber fluidly coupled to the second process volume.

[0006] In yet another embodiment, a processing system for semiconductor manufacturing is

provided. The processing system includes a chamber housing, a first process chamber, a second process chamber, a pressure sensor, and a foreline disposed in the chamber housing and between the first process chamber and the second process chamber. The first process chamber is disposed in the chamber housing and includes a first pump ring and a first liner. The first pump ring is a metal pump ring and includes a first port, a first baffle, and a first pressure conduit disposed in the first pump ring. The first pump ring and first liner partially define a first process volume. The second process chamber is in the chamber housing and includes a second pump ring and a second liner. The second pump ring is a metal pump ring and includes a second port, a second baffle, and a second pressure conduit disposed in the second pump ring. The second pump ring and second liner partially define a second process volume. The pressure sensor is in fluid communication with the first process volume by the first pressure conduit, the chamber housing disposed at least partially between the pressure sensor and the first pressure conduit of the first pump ring. The foreline includes a first exhaust chamber fluidly coupled to the first process volume and a second exhaust chamber fluidly coupled to the second process volume.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrate only exemplary embodiments and are therefore not to be considered limiting of the scope of the disclosure, as the disclosure may admit to other equally effective embodiments.

[0008] FIG. **1** shows a top plan view of an exemplary processing system according to some embodiments.

[0009] FIG. **2** shows a schematic cross-sectional view of an exemplary plasma system according to some embodiments.

[0010] FIG. **3** shows a schematic cross-sectional view of an exemplary processing chamber according to some embodiments.

[0011] FIG. **4** shows a schematic cross-sectional view of an exemplary processing chamber according to some embodiments.

[0012] FIG. **5** shows a schematic cross-sectional view of an exemplary processing chamber according to some embodiments.

[0013] FIG. **6** shows an isometric view of an exemplary pump ring assembly according to some embodiments.

[0014] FIG. **7** shows an isometric view of an exemplary liner according to some embodiments.

[0015] FIG. **8** shows an isometric view of an exemplary pump ring according to some embodiments.

[0016] To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

[0017] Plasma enhanced deposition processes may energize one or more constituent precursors to facilitate film formation on a substrate. Any number of material films may be produced to develop semiconductor structures, including conductive and dielectric films, as well as films to facilitate transfer and removal of materials. For example, hardmask films may be formed to facilitate patterning of a substrate, while protecting the underlying materials to be otherwise maintained. In many processing chambers, a number of precursors may be mixed in a gas panel and delivered to a processing region of a chamber where a substrate may be disposed. While components of the lid stack may impact flow distribution into the processing chamber, many other process variables may similarly impact uniformity of deposition.

[0018] As device features reduce in size, tolerances across a substrate surface may be reduced, and material property differences across a film may affect device realization and uniformity.

[0019] Many chambers include a characteristic process signature, which may produce residual non-uniformity across a substrate. Temperature differences, flow pattern uniformity, and other aspects of processing may impact the films on the substrate, creating film uniformity differences across the substrate for materials produced or removed. For example, turbulent deposition gas flow and/or misalignment of apertures of a blocker plate and faceplate of a gas box may lead to non-uniform flow of deposition gases. In some instances, the blocker plate may not uniformly distribute flow of precursors to edge regions of a substrate. Additionally, in one embodiment which may be combined with other embodiments a substrate support or heater on which a substrate is disposed may include one or more heating mechanisms to heat a substrate. When heat is delivered or lost differently between regions of a substrate, the film deposition may be impacted where, for example, warmer portions of the substrate may be characterized by thicker deposition or different film properties relative to cooler portions. This temperature non-uniformity may be attributable, for example, to temperature fluctuations about the shaft of the pedestal and may particularly affect edge regions of substrates.

[0020] Further, separate chambers' performing similar processes on similar substrates potentially introduces unknown variables that can reduce throughput. In contrast, by increasing the number of substrates a system with uniform chambers can process, throughput is enhanced by the reduction of potentially introduced unknown variables.

[0021] The present technology overcomes these challenges by using processing chamber designs that provide symmetry within the processing environment. For example, the chambers described herein may provide symmetric gas flow, thermal distribution, and RF distribution, which may improve deposition characteristics. Additionally, the chamber designs may include processing regions that are isolated from the transfer region and cooler components, which may reduce the need for use of purge gases during processing operations and may therefore reduce dilution of process fluids, which may increase the material efficiency of the chamber. Additionally, the isolation of the processing region from the transfer region may enable a number of processing chambers within a single system to be operated independently of one another. Embodiments may also utilize non-motorized lift pins, which may enable a simplified design that eliminates the need for vertical motion of a transfer robot to handoff substrates with the substrate support. Accordingly, the present technology may produce improved film deposition characterized by improved thickness and material property uniformity across a surface of the substrate.

[0022] Although the remaining disclosure will routinely identify specific deposition processes utilizing the disclosed technology, it will be readily understood that the systems and methods are equally applicable to other deposition and cleaning chambers, as well as processes as may occur in the described chambers. Accordingly, the technology should not be considered as limited as for use with these specific deposition processes or chambers alone.

[0023] The disclosure discusses a possible system and chamber that may include lid stack components according to embodiments of the present technology before additional variations and adjustments to this system according to embodiments of the present technology are described.

[0024] The system increases throughput by reducing the size of the process volume in process chambers and a shared foreline enables about equal pressure in the process volumes that are supplied by the foreline. Further, the system is able to independently monitor the pressure in each process volume by pressure conduits that connect a process volume to a specific pressure sensor. Each process chamber also includes a baffle within a pump ring assembly. The baffle enhances deposition uniformity by distributing process fluids radially within the pump ring assembly before the process fluid passes through a liner and into a process volume.

[0025] FIG. **1** shows a top plan view of one embodiment of a processing system **100** of process chambers according to embodiments. In the figure, a pair of front opening unified pods **102** supply substrates of a variety of sizes that are received by robotic arms **104** and placed into a low pressure holding area **106** before being placed into one of the substrate process chambers **108***a-f*, positioned in tandem sections **109***a-c*. A second robotic arm **110** may be used to transport the substrate wafers from the holding area **106** to the substrate process chambers **108***a-f* and back. Each substrate process chamber **108***a-f*, can be outfitted to perform a number of substrate processing operations including formation of stacks of semiconductor materials described herein in addition to plasma-enhanced chemical vapor deposition, atomic layer deposition, physical vapor deposition, etch, pre-clean, degas, orientation, and other substrate processes including, annealing, ashing, etc.

[0026] The substrate process chambers **108***a-f* may include one or more system components for depositing, annealing, curing and/or etching a dielectric or other film on the substrate. In one configuration, two pairs of the processing chambers, e.g., **108***c-d* and **108***e-f*, may be used to deposit dielectric material on the substrate, and the third pair of processing chambers, e.g., **108***a-b*, may be used to etch the deposited dielectric. In another configuration, all three pairs of chambers, e.g., **108***a-f*, may be configured to deposit stacks of alternating dielectric films on the substrate. Any one or more of the processes described may be carried out in chambers separated from the fabrication system shown in different embodiments. It will be appreciated that additional configurations of deposition, etching, annealing, and curing chambers for dielectric films are contemplated by the processing system **100**.

[0027] As shown, a controller **190** is in communication with the processing system **100** and is used to control processes and methods, such as the operations of the methods described herein. The controller **190** is configured to receive data or input as sensor readings from sensor(s) (such as one or more of sensor devices **281**, **282** discussed below). The sensor devices can include, for example: sensor devices that monitor pressure in the processing chambers, e.g., **108***a-b*.

[0028] The controller **190** includes a central processing unit (CPU) **193** (e.g., a processor), a memory **191** containing instructions, and support circuits **192** for the CPU **193**. The controller **190** controls various items directly, or via other computers and/or controllers. In one embodiment which can be combined with other embodiments, the controller **190** is communicatively coupled to dedicated controllers, and the controller **190** functions as a central controller.

[0029] The controller **190** is of any form of a general-purpose computer processor that is used in an industrial setting for controlling various substrate processing chambers and equipment, and sub-processors thereon or therein. The memory **191**, or non-transitory computer readable medium, is one or more of a readily available memory such as random access memory (RAM), dynamic random access memory (DRAM), static RAM (SRAM), and synchronous dynamic RAM (SDRAM (e.g., DDR1, DDR2, DDR3, DDR3L, LPDDR3, DDR4, LPDDR4, and the like)), read only memory (ROM), floppy disk, hard disk, flash drive, or any other form of digital storage, local or remote. The support circuits **192** of the controller **190** are coupled to the CPU **193** for supporting the CPU **193**. The support circuits **192** include cache, power supplies, clock circuits, input/output circuitry and subsystems, and the like.

[0030] The controller **190** is configured to conduct any of the operations described herein. The instructions stored on the memory, when executed by the CPU **193**, cause one or more of the operations described herein to be conducted in relation to the processing system **100**. The controller **190** and the processing system **100** are at least part of a system for processing substrates.

[0031] The various operations described herein can be conducted automatically using the controller **190**, or can be conducted automatically or manually with certain operations conducted by a user.

[0032] FIG. **2** shows a schematic cross-sectional view of an exemplary plasma system **200** according to one embodiment which may be combined with other embodiments.

[0033] The plasma system **200** is a tandem section **109**(*a-c*) of FIG. **1**. The plasma system **200** includes a foreline **201**, the first process chamber **108***a*, and the second process chamber **108***b*.
[0034] The process chambers **108***a*, **108***b* are disposed in a chamber housing **205**. In some embodiments that may be combined with other embodiments, the chamber housing **205** is a metallic monolithic chamber housing.
[0035] The first process chamber **108***a* includes a first process volume **217**, a showerhead **219**, a seal plate **214**, a substrate support **215**, a first pump ring assembly **213**, a spacer **251**, a lift column **241**, and a lid **253**. A substrate **216** is shown positioned on the substrate support **215**.
[0036] The showerhead **219**, the first pump ring assembly **213**, and substrate support define the first process volume **217**.
[0037] The first pump ring assembly **213** is in fluid communication with the foreline **201** by a first exhaust chamber **211**.
[0038] The second process chamber **108***b* includes a second process volume **237**, a showerhead **239**, a seal plate **234**, a substrate support **235**, a second pump ring assembly **233**, a spacer **251**, a lift column **243**, and a lid **253**. In some embodiments, the lid **253** is a single piece. The spacer **251** is a ceramic. The lid **253** is a metal alloy, for example, an aluminum alloy. The spacer **251** separates the lid **253** from the showerheads **219**, **239** of the first and second process chambers **108***a*, **108***b*. In some embodiments, each of the first and second process chambers **108***a*, **108***b* has its own spacer **251**. A substrate **236** is shown positioned on the substrate support **235**.
[0039] The showerhead **239**, the second pump ring assembly **233**, and the substrate support **235** define the second process volume **237**.
[0040] The second pump ring assembly **233** is in fluid communication with the foreline **201** by a second exhaust chamber **231**.
[0041] The foreline **201** includes a foreline chamber **203**, the first exhaust chamber **211**, and the second exhaust chamber **231**. The exhaust chambers **211**, **231** fluidly couple the first process volume **217** and the second process volume **237** to the foreline chamber **203** of the foreline **201**. The first exhaust chamber **211** and the second exhaust chamber **231** are offset from a major vertical axis A of the foreline chamber **203**.
[0042] The first exhaust chamber **211** is disposed between the foreline chamber **203** of the foreline **201** and the first process chamber **108***a*. The first exhaust chamber **211** separates the foreline **201** from the first process chamber **108***a*. The second exhaust chamber **231** is disposed between the foreline chamber **203** of the foreline **201** and the second process chamber **108***b*. The second exhaust chamber **231** separates the foreline **201** from the second process chamber **108***b*. The foreline **201**, the exhaust chambers **211**, **231**, and the foreline chamber **203** are disposed in the housing **205** and between the process chambers **108***a*, **108***b*.
[0043] The first process volume **217** is separated from the second process volume **237** by the first exhaust chamber **211**, the second exhaust chamber **231**, and the foreline **201**. The configuration of the foreline **201**, the first exhaust chamber **211**, and the second exhaust chamber **231**, is configured to enable pressure equalization between the first process volume **217** and the second process volume **237**.
[0044] The plasma system **200** is configured such that the first process volume **217** and the second process volume **237** are disposed in about the same plane. For example, the plane of the first process volume **217** and the second process volume **237** the foreline chamber **203** is disposed between the showerheads **219**, **239** and the foreline chamber **203**. In one embodiment which may be combined with other embodiments, the substrates **216**, **236** are disposed above the foreline chamber **203** and the exhaust chambers **211**, **231** during processing.
[0045] During processing, the pressure of the first process volume **217** and the second process volume **237** is between about 1 Torr to about 15 Torr, for example from about 2 Torr to about 10 Torr.
[0046] The lift columns **241**, **243** translate the seal plates **214**, **234**. When the seal plates **214**, **234**

are disposed in a process position, as show in in FIG. **2**, they form the process volumes **217**, **237**. When the seal plates **214**, **234** are disposed in a transfer position (not shown) the first seal plate **214** is disposed away from the first pump ring assembly **213** and the second seal plate **234** is disposed away from the second pump ring assembly **233**. This configuration enables a reduction in size of the process volumes **217**, **237**. In some embodiments, the lift columns **241**, **243** may each include an internal port to supply a purge gas to their respective process volumes **217**, **237**.

[0047] The first process volume **217** and the second process volume **237** are reduced volumes, for example, less than 5 liters. In some embodiments, which may be combined with other embodiments, the process volumes **217**, **237** are about 2 liters or less. For example, the process volumes **217**, **237** are about 0.5 liters to about 5 liters. In the process volumes **217**, **237** are about 2.5 liters to about 3.5 liters.

[0048] The shared foreline **201** and individual first exhaust chamber **211** and the second exhaust chamber **231** enable separate process volumes with about equal pressures without a dedicated equalization port between the first process volume **217** and the second process volume **237**. By reducing the size of the process volumes **217**, **237**, the shared foreline **201** is able to exhaust process fluid and keep the pressures in first process volume **217** and the second process volume **237** at about equal.

[0049] The shared foreline **201** is configured to exhaust the process fluid from the first process volume **217** and the second process volume **237**. The process fluid may include Tetraethyl Orthosilicate (TEOS) supplied at about 4,000 to about 6,000 milligrams per minute, oxygen at about 2,700 standard cubic centimeter per minute (sccm) to about 10,000 sccm, argon at about 3,200 sccm to about 8,000 sccm and any combination thereof. The process fluid may also include Silane, silane oxide, silane nitride, Ammonia

[0050] The plasma system **200** includes a first pressure sensor **281** and a second pressure sensor **282**. The first pressure sensor **281** and the second pressure sensor **282** read the pressure in the first process volume **217** and the second process volume **237** of the first process chamber **108***a*, and the second process chamber **108***b* respectively. The first pressure sensor **281** and the second pressure sensor **282** are monometers according to some embodiments. The pressure sensors **281**, **282** are in fluid communication with the corresponding process volumes through conduits in the pump ring assemblies **213**, **233** and housing **205**. The conduits are described in more detail with respect to FIG. **4**.

[0051] FIG. **3** shows a schematic cross-sectional view of an exemplary process chamber **108***a* according to one embodiment which may be combined with other embodiments. While the first process chamber **108***a* is shown, the second process chamber **108***b* may include the same or similar features. In one embodiment which may be combined with other embodiments, the first process chamber **108***a* is a mirror image of the second process chamber **108***b*.

[0052] The first pump ring assembly **213** includes a first liner **305**, a first ring housing **303**, and a first pump ring **301**.

[0053] The first pump ring **301** is a metal pump ring according to one embodiment which may be combined with other embodiments. The first pump ring **301** and first liner **305** partially define the first process volume **217**. The first process volume **217** may be further defined by the substrate support **215**, the first liner **305**, and the first showerhead **219**. The first showerhead **219** is a ceramic according to one embodiment which may be combined with other embodiments.

[0054] The first pump ring **301** includes a first port **311** and a first baffle **317**. The first pump ring **301**, the first baffle **317** of the first pump ring **301**, and the first ring housing **303** form a first port cavity **307**.

[0055] The first port cavity **307** separates the first port **311** from the first ring housing **303** and the first baffle **317**. The first port cavity **307** is fluidly coupled to the first exhaust chamber **211** by the first port **311**.

[0056] According to one embodiment which may be combined with other embodiments, the first seal plate **214** includes a heater **315** disposed within the seal plate **214**.

[0057] According to one embodiment which may be combined with other embodiments, the substrate support **215** includes the heater **315** disposed within the substrate support **215**.

[0058] The first liner **305** includes one or more first exhaust apertures **309**. The first exhaust apertures **309** are disposed radially through the first liner **305**. The first exhaust apertures **309** fluidly couple the first port cavity **307** to the first process volume **217**.

[0059] The first baffle **317** is disposed between the first port cavity **307** and the first exhaust apertures **309**.

[0060] FIG. **4** shows a schematic cross-sectional view of an exemplary processing chamber **400** according to some embodiments. The processing chamber **400** may be the process chamber **108***a* or the process chamber **108***b*.

[0061] The processing chamber **400** includes the housing **205**, a seal plate **414**, a substrate support **415**, a heater **412**, a pump ring **401**, a liner **405**, a ring housing **403**, a spacer **451**, one or more seals **437**, a lid **453**, and a showerhead **419**. A substrate **416** is shown positioned on the substrate support **415**.

[0062] At least some of the one or more seals **437** are disposed between the housing **205** and the pump ring **401**. At least one of the one or more seals **437** are disposed between the ring housing **403** and the lid **453**. The seals **437** may also be disposed between the spacer **451** and the lid **453** and also may be disposed between the spacer **451** and the showerhead **419**. The spacer **451** is disposed between the lid **453** and the showerhead **419**.

[0063] In one embodiment which may be combined with other embodiments, the spacer **451** and liner **405** are made of ceramic materials and are configured to protect the metallic lid **453**, ring housing **403**, and pump ring **401** from RF signals produced during a process. The spacer **451** and liner **405** insulate the showerhead **419** from the grounded lid **453** and chamber housing **205**.

[0064] The seal plate **414** has a sealing surface **435** that seals against the pump ring **401**. The sealing surface **435** extends radially outward of the substrate support **415**. The sealing surface **435** may include an elastomeric seal, a polymer seal, a radio frequency seal, or any combination thereof.

[0065] The sealing surface **435** of the seal plate **414** defines a lower bound of the process volume **417**. For example, during processing, process fluid is disposed between the showerhead **419** and the sealing surface **435** to for a reduced process volume **417**. The seal plate **414** is a metallic material and supports the substrate support **415**. The substrate support **415** is a ceramic support disposed on the seal plate **414** away from the portion of the seal plate **414** from which the sealing surface **435** extends.

[0066] The pump ring **401** includes a pressure conduit **431** disposed in the pump ring **401**. The pressure conduit **431** of the pump ring **401** is in fluid communication with a pressure conduit **433** of the housing **205**. The pressure conduits **431**, **433** connect to a pressure sensor. For example the pressure conduits **431**, **433** fluidly couple a pressure sensor **282** (FIGS. **2**, **3**) to the process volume **417**. For example, the pressure conduits **431**, **433** connect the process volume **417** to the first pressure sensor **281** and/or the second pressure sensor **282** (FIG. **2**). The pressure conduits **431**, **433** have a diameter between about 3 millimeters and about 5 millimeters.

[0067] The pressure conduit **431** is disposed proximate the plane of the substrate **416** and process volume **417**. The pressure conduit **431** is disposed between the sealing surface **435** of the seal plate **414** and the showerhead **419**. For example, the pressure conduit **431** is disposed at about 10 millimeters to about 30 millimeters from the plane of the substrate **416** and process volume **417**. The pressure conduits **431**, **433** allow for the pressure sensor **282** (FIG. **3**) to experience a pressure measurement adjacent and directly connected to the process volume **417**. The proximity of the pressure conduit **431** allows for a reduction in potential sources of inaccuracy.

[0068] The liner **405**, the pump ring **401**, a baffle **410**, and the ring housing **403** form an exhaust cavity **411**. Process fluid leaves from the process volume **417** to the exhaust cavity **411** through one or more exhaust apertures **409** of the liner **405**. For example, the liner **405** has about 100 to 200 exhaust apertures **409** in a radial array circumscribing the process volume **417** and through the liner **405**. Process fluid from the process volume **417** applies pressure to a fluid in the pressure conduits **431**, **433** which enables localized pressure measurement of the process volume **417** by the pressure sensor **282** (FIG. **3**). The localized pressure measurement enhances measurement accuracy of the processing chamber **400**.

[0069] FIG. **5** shows a schematic cross-sectional view of an exemplary processing chamber **500** according to some embodiments. The processing chamber **500** may be the process chamber **108***a*, the process chamber **108***b*, or the processing chamber **400**.

[0070] The processing chamber **500** includes a port cavity **407** in fluid communication with the foreline **201** by an exhaust chamber **531**. The exhaust chamber **531** is in fluid communication with the port cavity **407** by a port **533**. Process fluid passes from the port cavity **407** to the foreline chamber **203** of the foreline **201**, through the port **533**.

[0071] The port cavity **407** is defined by the baffle **410**, the pump ring **401**, the ring housing **403** and the port **533**. Process fluid flows from the exhaust cavity **411** around the baffle **410** and between the pump ring **401** and the ring housing **403** to enter the port cavity **407**. Process fluid flows out of the port cavity **407** through the port **533** to the exhaust chamber **531**. Process fluid leaves the process volume **417** to the exhaust cavity **411** through the exhaust apertures **409**.

[0072] FIG. **6** shows an isometric view of an exemplary pump ring assembly **600** according to some embodiments. The pump ring assembly **600** may be the pump ring assembly **213** or pump ring assembly **233** of FIGS. **2** and **3**.

[0073] The pump ring assembly **600** includes the ring housing **403**, the pump ring **401**, and the liner **405**. The liner **405** is illustrated away from the ring housing **403** for illustrative purposes. The liner **405** is disposed radially inward of the ring housing **403**.

[0074] When the liner **405** is placed within the pump ring assembly **600**, the pump ring **401**, the ring housing **403**, and the liner **405** form the exhaust cavity **411**.

[0075] The ring housing **403** includes mounting holes **601**, a tongue **603**, and seals **437**. The mounting holes **601** are disposed in a radial array around the process volume **417**. The mounting holes **601** are configured to allow the pump ring assembly **600** to be coupled to the housing **205** (FIG. **2**).

[0076] The tongue **603** is the portion of the ring housing **403** that is disposed over the port **533** and partially defines the port cavity **407** (FIG. **5**). In one embodiment which may be combined with other embodiments the tongue **603** is a trapezoidal feature of the ring housing **403**.

[0077] FIG. **7** shows an isometric view of an exemplary liner **405** according to some embodiments.

[0078] The liner **405** includes an inner surface **703**, the exhaust apertures **409**, a cavity surface **707**, an extension **711**, and an index feature **705**.

[0079] The extension **711** is a portion of the liner **405** that extends from the inner surface **703** radially outward past the cavity surface **707**.

[0080] The index feature **705** is a semicircular notch in the liner **405** that is configured to align with pressure conduit **431**. The pressure conduit **431** has geometry that is discussed below.

[0081] The index feature **705** and the exhaust apertures **409** are disposed from the inner surface **703** radially outward past the cavity surface **707**. The exhaust apertures **409** are disposed between the index feature **705** and the extension **711**. The outer radial face of the extension **711** is disposed in contact and seals with an inner radial face of the ring housing **403** (FIG. **6**).

[0082] FIG. **8** shows an isometric view of an exemplary pump ring **401** according to some embodiments.

[0083] The pump ring **401** includes an inner surface **801**, an inner radial surface **803**, a top surface **805**, a conduit feature **809**, and a wall **807**.

[0084] The inner surface **801** is about perpendicular to the inner radial surface **803**. The port **533** is an aperture through the inner surface **801**. The inner radial surface **803** circumscribes and faces the process volume **417**.

[0085] The wall **807** is the outer most body of the pump ring **401**. In one embodiment which may be combined with other embodiments the wall **807** and the baffle **410** extend from the inner surface **801** at an about equal length. For example, the baffle **410** extends from the inner surface **801** from about 1 inch to about 1.5 inches, for example about 1.2 inches.

[0086] The baffle **410** extends from the inner surface **801** to about parallel with the top surface **805**. The top surface **805** and the baffle **410** seal against the ring housing **403** (FIG. **6**). The port cavity **407** is partially defined by the wall **807**, the inner surface **801**, and a second section **812** of the baffle **410**. The baffle **410** partially circumscribes the process volume **417**.

[0087] The baffle **410** includes one or more sections separated by slits. In one embodiment which may be combined with other embodiments, the baffle **410** includes a first section **811**, the second section **812**, and a third section **813**. The first and second sections **811**, **812** are separated by a first slit **821**. The second and third sections **812**, **813** are separated by a second slit **823**. In one embodiment which may be combined with other embodiments the first section **811**, the second section **812**, and the third section **813** are about equal. In one embodiment which may be combined with other embodiments the first section **811** and the third section **813** are larger than the second section **812**. For example, an arc length of the first section **811** and the third section **813** is about 10% or more than the arc length of the second section **812**.

[0088] In one embodiment which may be combined with other embodiments the first section **811** and the third section **813** are smaller than the second section **812**. For example, an arc length of the second section **812** is about 10% or more than the arc lengths of the first section **811** and the third section **813**.

[0089] The first slit **821** is the gap between the first and second sections **811**, **812**. The second slit **823** is the gap between the second and third sections **812**, **813**. The first slit **821** and the second slit **823** are sized based on the process fluid used. For example, the first slit **821** and the second slit **823** are between about 2 millimeters and about 4 millimeters.

[0090] In one embodiment which may be combined with other embodiments the baffle **410** is disposed radially around about half or more of the process volume **417**. For example, the baffle **410** is disposed circumferentially around 40% to about 60% of the process volume **417**. In one embodiment which may be combined with other embodiments the baffle **410** is disposed in adjacent segments around a portion of the process volume **417**. The baffle **410** is disposed at an offset from the inner radial surface **803** to partially form the exhaust cavity **411**.

[0091] The conduit feature **809** is a raised portion of the inner surface **801** of the pump ring **401**. The conduit feature **809** is concentric with the pressure conduit **431**. The conduit feature **809** extends through the baffle **410**. The conduit feature **809** is disposed between the first slit **821** and the second slit **823**. The conduit feature **809** extends from the process volume **417**, through the baffle **410**, to the port cavity **407**. The conduit feature **809** allows the pressure conduit **431** to be disposed closer to the process volume **417** from the sealing surface **435** of the seal plate **414** (FIG. **4**). Disposing the pressure conduit **431** closer to the process volume **417** allows the pressure conduit **431** to experience the pressure of the process volume **417** with enhanced accuracy. The pressure conduit **431** is an aperture through the inner radial surface **803** that allows the pressure sensor **282** (FIG. **3**) to experience the process volume **417**.

[0092] The conduit feature **809** indexes with the index feature **705** which allows the pressure conduit **431** to be disposed elevated in relation to the inner surface **801** such that the pressure conduit **431** is closer to the top surface **805**

[0093] The exhaust cavity **411** is defined by the baffle **410**, the portion of the wall **807** across the process volume **417** from the baffle **410**, and the portion of the inner surface **801** radially inward of the baffle **410**.

[0094] As process fluid travels from the port **533**, the process fluid leaves the port cavity **407**. Process fluid enters the exhaust cavity **411** through the exhaust apertures **409**. Process fluid travels through the slits **821**, **823** and around the distal edges of the first and third sections **811**, **813**, as it leaves the exhaust cavity **411** and enters the port cavity **407**. The baffle **410** allows for enhanced flow by equalizing the flow rate of process fluid through exhaust apertures **409**.

[0095] During a process the flow of process fluid is in the incompressible flow regime. For example, the first slit **821** and the second slit **823** allow for flow of process fluid in the viscous flow regime and enhance deposition uniformity during a process.

[0096] The slits **821**, **823** of the baffle **410** represent about 10% or less of the baffle **410**. For example, in embodiments where the baffle **410** does not have a slit, the baffle is at 100%. In some embodiments that may be combined with other embodiments, the slits **821**, **823** allow process fluid to pass through 10% or less of the baffle.

[0097] The baffle **410** and slits **821**, **823** enhance azimuthal flow of process fluid in the process volume **417** by exhausting the process fluid through the exhaust apertures **409** of the liner **405**. The azimuthal flow of process fluid in the process volume **417** enhances deposition uniformity on a substrate.

[0098] Benefits of the present disclosure include reduced gas consumption and gas waste; increased chamber condition measurement accuracy; and more uniform deposition. Benefits also include enhanced manufacturability, reduced divertive gas flow, and reduction of processing volume size.

[0099] It is contemplated that one or more aspects disclosed herein may be combined. As an example, one or more aspects, features, components, operations and/or properties of the process chambers **108***a*, **108***b;* the controller **190**; the one or more sensor devices **281**, **282**; the baffle **410**; the foreline **201**; the pressure conduit **431**, **433**; and/or the exhaust chambers **211**, **231** may be combined to enhance measurement accuracy, through put, and or deposition uniformity.

[0100] While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow

Claims

1. A processing system for semiconductor manufacturing, the processing system comprising: a chamber housing; a first process chamber disposed in the chamber housing, the first process chamber comprising: a first liner; and a first pump ring, the first pump ring comprising: a first port; and a first baffle, the first pump ring and first liner partially defining a first process volume; a second process chamber disposed in the chamber housing, the second process chamber comprising: a second liner; and a second pump ring, the second pump ring comprising: a second port; and a second baffle, the second pump ring and second liner partially defining a second process volume; and a foreline disposed in the chamber housing and between the first process chamber and the second process chamber, the foreline comprising: a first exhaust chamber fluidly coupled to the first process volume; and a second exhaust chamber fluidly coupled to the second process volume.

2. The processing system of claim 1, wherein the first liner, the first baffle, and the first pump ring partially define an exhaust cavity.

3. The processing system of claim 2, wherein the first liner comprises an exhaust aperture, the first baffle disposed between the first port and the exhaust aperture.

4. The processing system of claim 1, wherein the first baffle is disposed radially around half or less of the first process volume.

5. The processing system of claim 1, wherein the first baffle comprises one or more slits disposed between and fluidly connecting the first process volume and the first port, the one or more slits forming one or more sections of the first baffle.

6. The processing system of claim 5, wherein the one or more sections of the first baffle are about equal.

7. The processing system of claim 1, further comprising: a lid; and a first ring assembly, the first ring assembly defining an exhaust cavity, the first ring assembly comprising: the first liner; a first ring housing disposed radially outward of the first liner; and the first pump ring, the distribution ring and the first ring housing disposed between the lid and the first pump ring.
8. The processing system of claim 1, wherein foreline further comprises a foreline chamber disposed between the first exhaust chamber and the second exhaust chamber, the first process volume being separated from the second process volume by the first exhaust chamber, the foreline chamber, and the second exhaust chamber.
9. The processing system of claim 1, wherein the first baffle is disposed between the first port and the first process volume.
10. A processing system for semiconductor manufacturing, the processing system comprising: a chamber housing; a first process chamber disposed in the chamber housing, the first process chamber comprising: a first liner; and a first pump ring, the first pump ring comprising: a first port; a first baffle, the first pump ring and first liner partially defining a first process volume; and a first pressure conduit disposed in the first pump ring; a second process chamber disposed in the chamber housing, the second process chamber comprising: a second liner; and a second pump ring, the second pump ring comprising: a second port; a second baffle, the second pump ring and second liner partially defining a second process volume; and a second pressure conduit disposed in the second pump ring; a foreline disposed in the chamber housing and between the first process chamber and the second process chamber, the foreline comprising: a first exhaust chamber fluidly coupled to the first process volume; and a second exhaust chamber fluidly coupled to the second process volume; and a pressure sensor in fluid communication with the first process volume by the first pressure conduit, the chamber housing disposed at least partially between the pressure sensor and the first pressure conduit of the first pump ring.
11. The processing system of claim 10, wherein the first pressure conduit passes through the first baffle.
12. The processing system of claim 10, wherein the first liner comprises an index feature configured to enable the first pressure conduit to experience a pressure in the first process volume.
13. The processing system of claim 10, wherein the first process chamber further comprises a ring housing, the ring housing disposed radially outward of the first liner.
14. The processing system of claim 13, wherein the ring housing, the first baffle, and the first pump ring partially define a port cavity.
15. The processing system of claim 14, wherein the first liner, the first baffle, and the first pump ring partially define an exhaust cavity, the first baffle disposed between the exhaust cavity and the port cavity.
16. The processing system of claim 10, wherein the chamber housing comprises a pressure conduit disposed between the pressure sensor and the first pressure conduit.
17. The processing system of claim 10, wherein the first baffle comprises one or more slits disposed between and fluidly connecting the first process volume and the first port, the one or more slits forming one or more sections of the first baffle.
18. The processing system of claim 10, wherein the first process volume is separated from the second process volume by the first exhaust chamber, the second exhaust chamber, and the foreline, the first baffle and second baffle, the first exhaust chamber, the second exhaust chamber, and the foreline configured to enable pressure equalization between the first process volume and the second process volume.
19. A processing system for semiconductor manufacturing, the processing system comprising: a chamber housing; a first process chamber disposed in the chamber housing, the first process chamber comprising: a first liner; and a first pump ring, the first pump ring being a metal pump ring, the first pump ring comprising: a first port; a first baffle, the first pump ring and first liner partially defining a first process volume, the first process volume being less than about 5 liters; and

a first pressure conduit disposed in the first pump ring; a second process chamber disposed in the chamber housing, the second process chamber comprising: a second liner; and a second pump ring, the second pump ring being a metal pump ring, the second pump ring comprising: a second port; a second baffle, the second pump ring and second liner partially defining a second process volume, the first process volume about equal to the second process volume; and a second pressure conduit disposed in the second pump ring; a foreline disposed in the chamber housing and between the first process chamber and the second process chamber, the foreline comprising: a first exhaust chamber fluidly coupled to the first process volume; and a second exhaust chamber fluidly coupled to the second process volume; and a pressure sensor in fluid communication with the first process volume by the first pressure conduit, the chamber housing disposed at least partially between the pressure sensor and the first pressure conduit of the first pump ring.

20. The processing system of claim 19, wherein the first process volume and the second process volume are in about a same plane.